Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 33)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 . 1 Compare selective breeding, asexual reproduction and cross-breeding as methods of improving livestock or crops.

[6 marks]

0 1 . 2 Explain how artificial insemination and embryo transfer increase the rate at which desirable characteristics spread through a livestock population.

[4 marks]

Do not write outside the box

10

0 2 . 1 State the Brundtland Commission (1987) definition of sustainable development.

[1 mark]

.....

.....

0 2 . 2 Explain why activities such as slash-and-burn agriculture, fishing and sewage discharge can be sustainable with a small human population but not with a large one.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 2 . 3 Explain why urban populations are often less aware of unsustainable exploitation than rural communities.

[2 marks]

.....

.....

.....

.....

0 2 . 4 Describe **two** negative feedback mechanisms in natural systems and explain how human responses can override them.

[4 marks]

.....

.....

.....

.....

.....

Do not write outside the box

Do not write outside the box

10

0 3 . 1 Explain how living organisms provide each of the following ecosystem services.

[6 marks]

Regulation of atmospheric composition

Soil maintenance

Interspecies relationships

0 3 . 2 Explain why Darwin's orchid of Madagascar depends on a single moth species, and why such specialisation is both an advantage and a risk.

[2 marks]

0 3 . 3 Explain why elephants are particularly important seed dispersers.

[2 marks]

.....

.....

.....

10

0 4 Table 1 shows some of the IUCN criteria used to place species in Red List categories.

Table 1

Category	Population decline over 10 years or 3 generations	Area of habitat	Number of mature adults	Probability of extinction in the wild
Vulnerable	over 40%	under 2000 km^2	under 10 000	over 10% in 100 years
Endangered	over 60%	under 500 km^2	under 2500	over 20% in 20 years or 5 generations
Critically endangered	over 80%	under 10 km^2	under 250	over 50% in 10 years or 3 generations

0 4 . 1 State **three** roles of the IUCN.

[3 marks]

1 .....

.....

2 .....

.....

3 .....

.....

0 4 . 2 A species of lemur has declined from 12 000 to 2 100 mature adults in three generations and occupies 400 km^2 of forest. Use Table 1 to determine its Red List category. Explain your answer.

[3 marks]

.....

.....

.....

.....

Do not write outside the box

Do not write outside the box

.....

.....

.....

0 4 . 3 Explain why only a small proportion of species have been assessed for the Red List, and give **two** criteria used to decide which species to assess.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 4 . 4 Give **one** example of a species whose Red List category improved because of conservation and state the actions responsible.

[1 mark]

.....

.....

10

0 5 . 1 Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

A Radio collar ☐

B Acoustic / satellite tag ☐

C Pitfall trap ☐

D Geolocator tag recovered by recapture ☐

0 5 . 2 Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

.....

.....

.....

Do not write outside the box

5

0 6 Table 1 shows three investigations carried out by a class.

Table 1

Investigation	Data collected
A	Mean height of bluebells in a coppiced and an uncoppiced wood (both normally distributed)
B	Number of limpets in each of five zones on a rocky shore, compared with the numbers expected if they were evenly spread
C	Soil moisture and the percentage cover of a moss, measured at 14 points along a transect

0 6 . 1 Name the most suitable statistical test for each investigation in Table 1.

[3 marks]

0 6 . 2 Explain what it means if a test gives a probability of $p = 0.02$.

[2 marks]

0 6 . 3 Explain why the calculated value from a test must be compared with a critical value, and what else is needed to find that critical value.

[3 marks]

0 6 . 4 Explain what the **standard deviation** of each set of bluebell heights tells the student, and how it is shown on a graph of the means.

[3 marks]

0 6 . 5 The class found a significant correlation in investigation C. Explain **four** reasons why this does not show that soil moisture causes the moss to grow.

[4 marks]

Do not write outside the box

Do not write outside the box

15

0 7 Table 1 shows climate data for two sites where a farmer is considering growing maize, a frost-sensitive crop that needs a thermal growing season of at least 120 days.

Table 1

Feature	Site A: valley floor	Site B: south-facing slope
Frost-free period / days	110	150
Days with mean temperature above 5 °C	160	175
Mean summer rainfall / mm	180	150
Gradient / %	1	8

0 7 . 1 Use Table 1 to explain which site is more suitable for maize.

[3 marks]

0 7 . 2 Describe **three** ways in which temperature can be controlled to increase crop productivity.

[3 marks]

1

2

3

0 7 . 3 Explain why keeping livestock warm increases the amount of food converted to growth.

[2 marks]

.....

.....

.....

.....

0 7 . 4 Explain how day length affects livestock production and how it can be manipulated.

[2 marks]

.....

.....

.....

.....

10

0 8 . 1 Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

A Close planting of a single non-indigenous species ☐

B Mixed indigenous species with a mixed age structure ☐

C Clear-felling on a 40-year rotation ☐

D Removal of dead wood ☐

0 8 . 2 Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

Do not write outside the box

1 .....

.....

2 .....

.....

3 .....

.....

4 .....

.....

5

0 9 . 1 Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

[1 mark]

A Its mass ☐

B Its magnetic field ☐

C Its distance from the Sun ☐

D Its speed of rotation ☐

0 9 . 2 Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

.....

.....

.....

.....

0 9 . 3 Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

.....

.....

.....

.....

Do not write outside the box

5

1 0 . 1 Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

1 0 . 2 Explain why growing coral reefs help to resist climate change.

[2 marks]

1 0 . 3 Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

Do not write outside the box

Do not write outside the box

Extra space

.....

.....

.....

.....

.....

15

Do not write outside the box

Write an essay on one of the following topics.

1 1 . 1 Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

OR

1 1 . 2 Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

END OF QUESTIONS